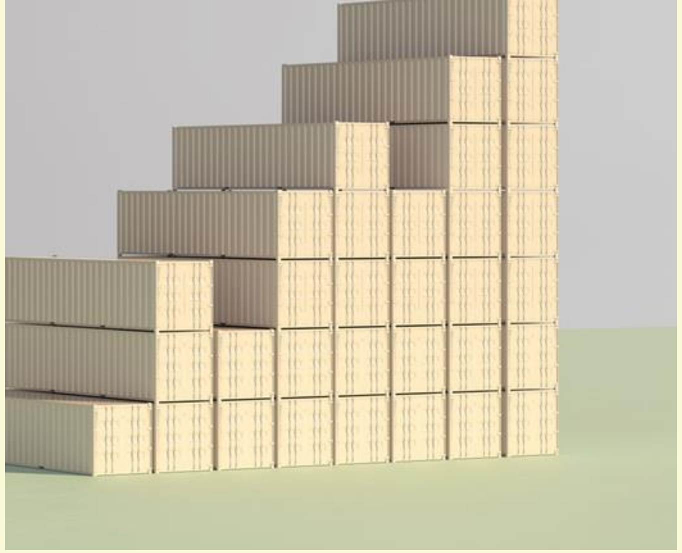


PROJECT 3A: VISUALIZATION OF TIME SERIES DATA

Data Analytics for FinTech
IDC4252C-2032
Karah Culley

Data Preparation

- The stock.csv file was uploaded into SAS Studio and imported into a WORK dataset
- The data was verified using PROC CONTENTS to ensure all variables were properly loaded

The output:

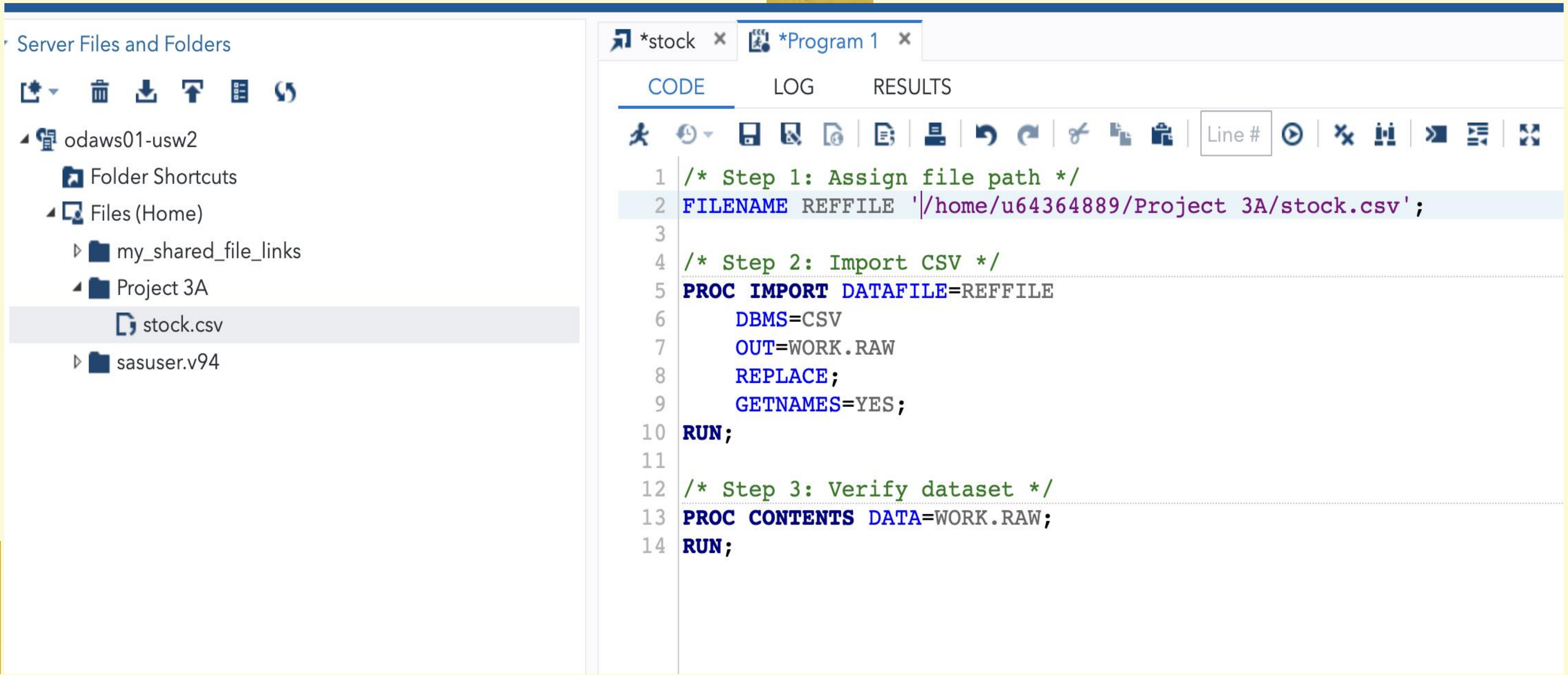
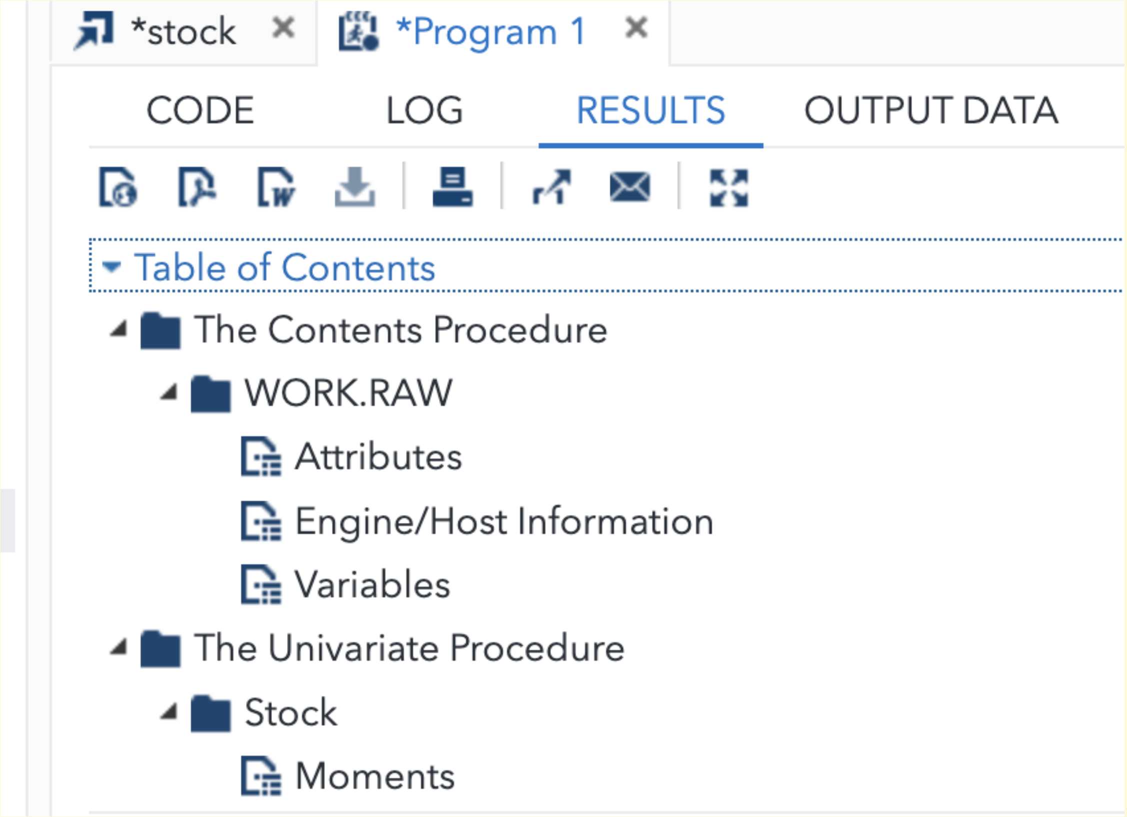
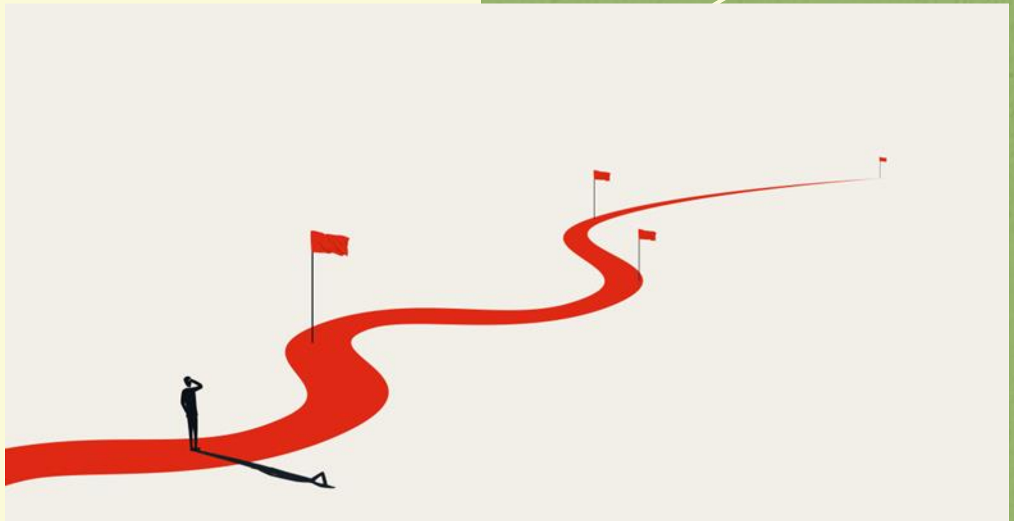


Table of Contents

This is the table of contents created for the SAS project:





Descriptive Statistics

The output:

- PROC UNIVARIATE was used to summarize the stock variable

 The output provides the measures:

  - Mean
  - Standard deviation
  - Minimum and maximum values
  - Quartiles

- These statistics help describe the overall distribution of stock prices

Tests for Location: Mu0=0

Test	Statistic		p Value	
Student's t	t	180.3643	Pr > |t|	<.0001
Sign	M	297	Pr >= |M|	<.0001
Signed Rank	S	88357.5	Pr >= |S|	<.0001

Quantiles (Definition 5)

Level	Quantile
100% Max	5.83
99%	5.81
95%	5.73
90%	5.63
75% Q3	5.44
50% Median	4.71
25% Q1	4.42
10%	4.07
5%	3.48
1%	3.44
0% Min	0.37

Extreme Observations

Lowest		Highest	
Value	Obs	Value	Obs
0.37	203	5.81	585
3.43	30	5.81	588
3.43	27	5.82	586
3.43	24	5.82	589
3.43	23	5.83	587

The UNIVARIATE Procedure
Variable: Stock

Moments

N	594	Sum Weights	594
Mean	4.88314815	Sum Observations	2900.59
Std Deviation	0.65984593	Variance	0.43539665
Skewness	-0.8858098	Kurtosis	2.62474774
Uncorrected SS	14422.2009	Corrected SS	258.190213
Coeff Variation	13.5127157	Std Error Mean	0.02707381

Basic Statistical Measures

Location		Variability	
Mean	4.883148	Std Deviation	0.65985
Median	4.710000	Variance	0.43540
Mode	4.270000	Range	5.46000
		Interquartile Range	1.02000

Tests for Location: Mu0=0

Test	Statistic		p Value	
Student's t	t	180.3643	Pr > |t|	<.0001
Sign	M	297	Pr >= |M|	<.0001
Signed Rank	S	88357.5	Pr >= |S|	<.0001

Quantiles (Definition 5)

Level	Quantile
100% Max	5.83
99%	5.81
95%	5.73
90%	5.63
75% Q3	5.44

Summary Statistics

- PROC MEANS was used to calculate:
  - Number of observations
  - Mean
  - Standard deviation
  - Minimum and maximum

- This confirms the central tendency and spread of the stock prices

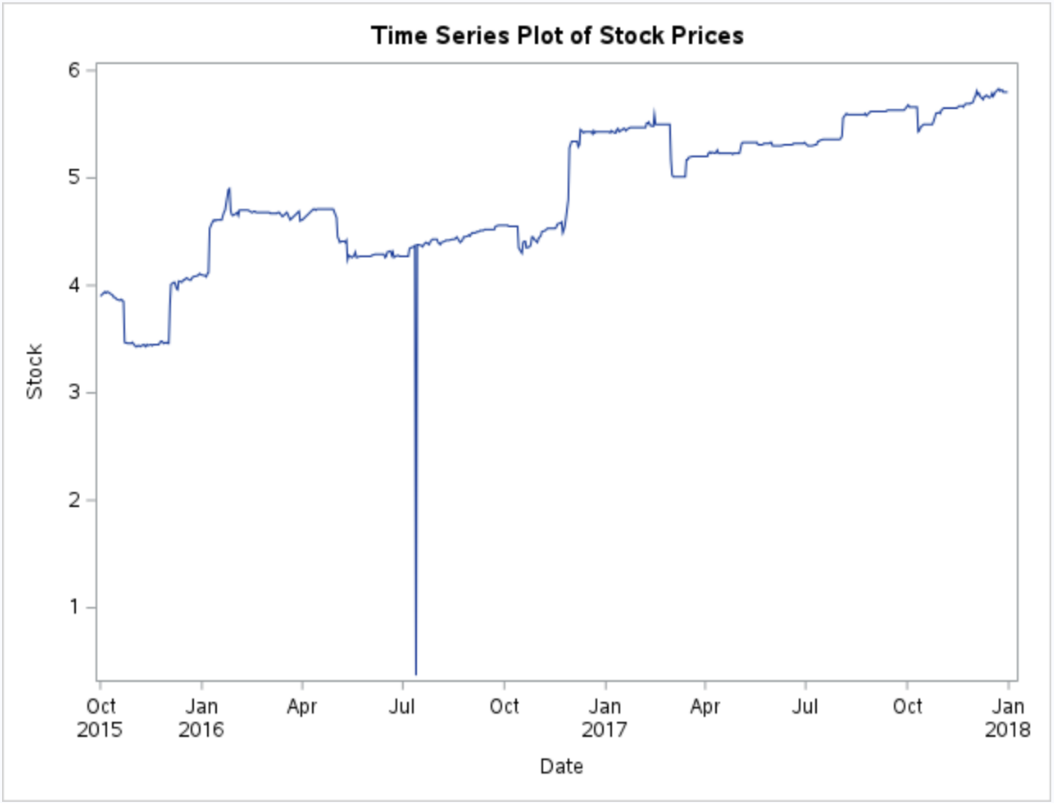
The output:

The MEANS Procedure

Analysis Variable : Stock				
N	Mean	Std Dev	Minimum	Maximum
594	4.8831481	0.6598459	0.3700000	5.8300000

Time Series Plot

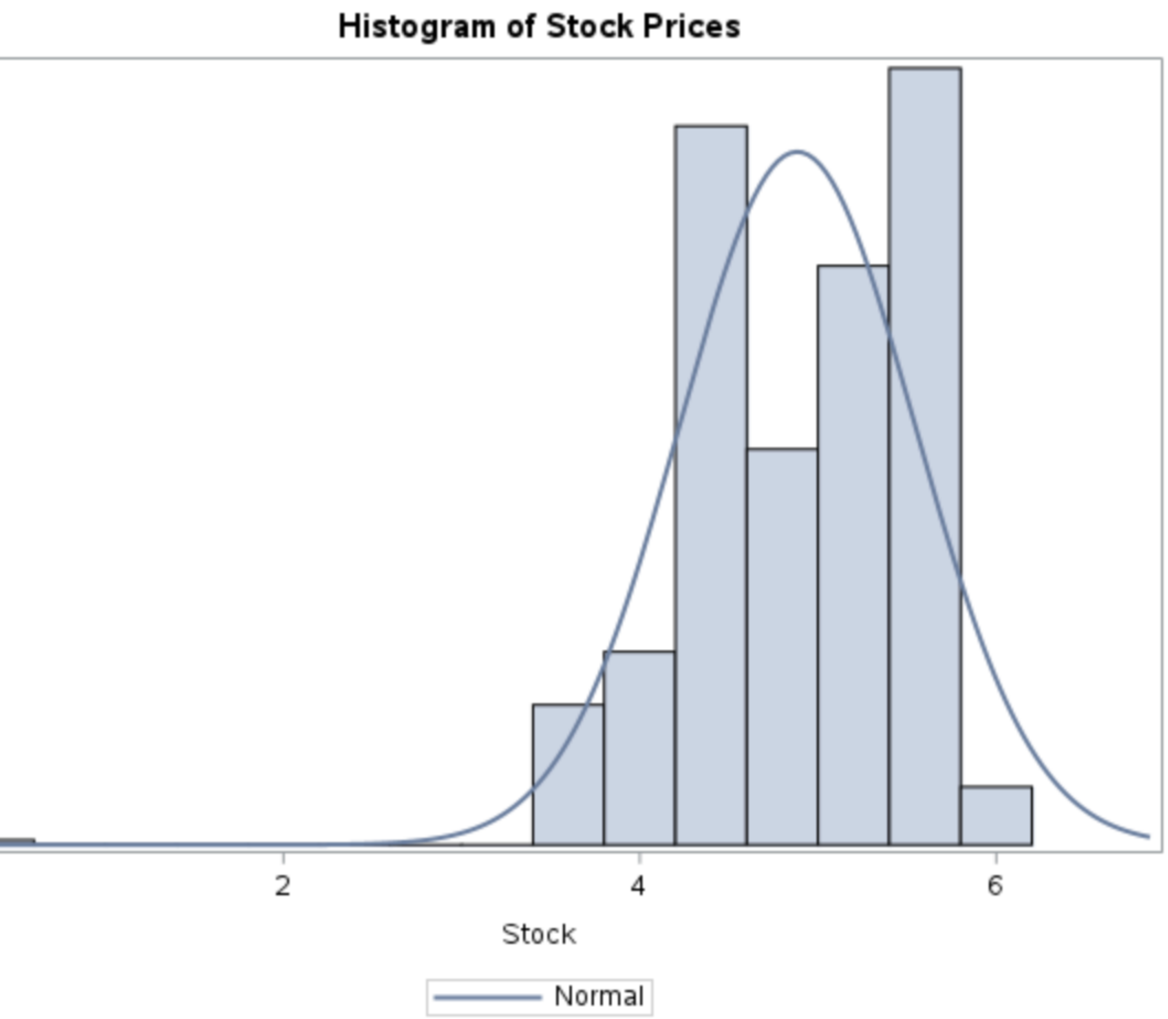
The output:



- A time series plot was created to show how the stock price changes over time

- The graph helps identify:
  - Overall trends
  - Periods of growth or decline
  - Volatility in the stock price

The output:



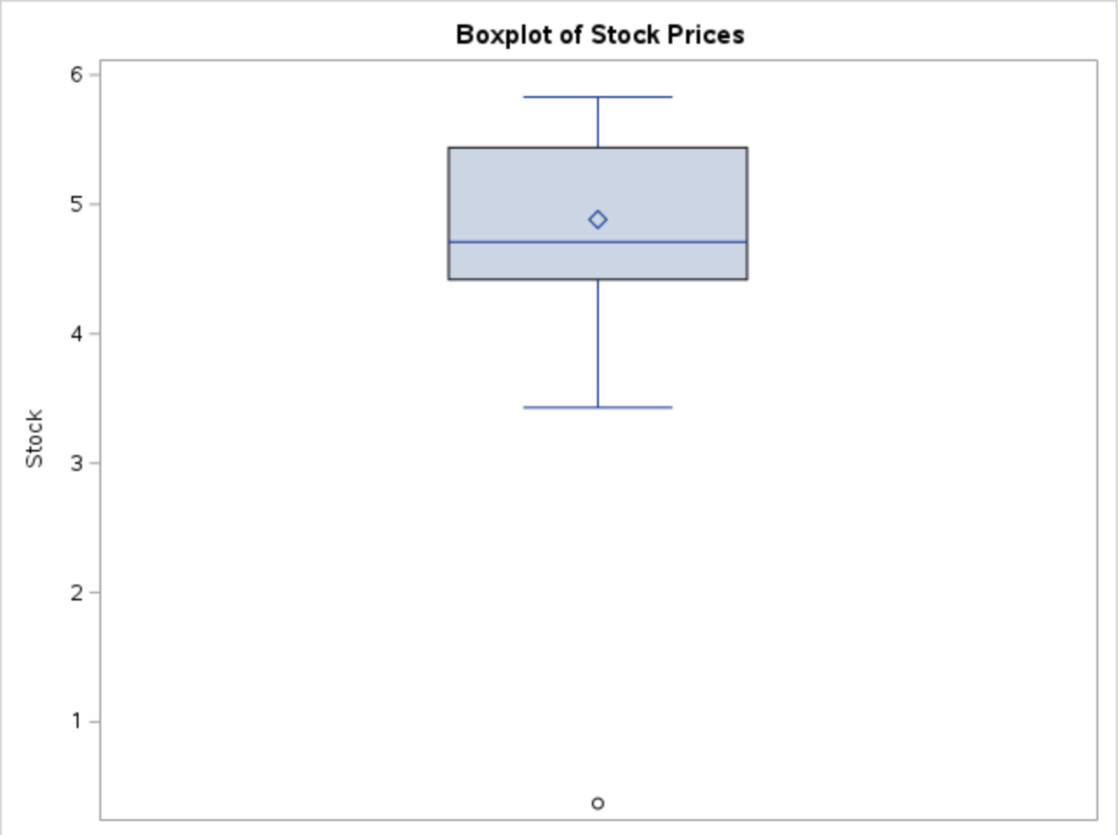
- A histogram was created to show the distribution of stock prices

- The graph shows:
  - How frequently different price levels occur
  - The overall shape of the distribution

DISTRIBUTION OF STOCK PRICES

BOXPLOT ANALYSIS

The output:



- A boxplot was used to visualize the spread of the stock prices.

- The boxplot shows:
  - The median
  - Interquartile range
  - Possible outliers

Correlation Analysis

- PROC CORR was used to measure the strength of relationships between stock price and independent variables
- Correlation values range from -1 to +1

The output:

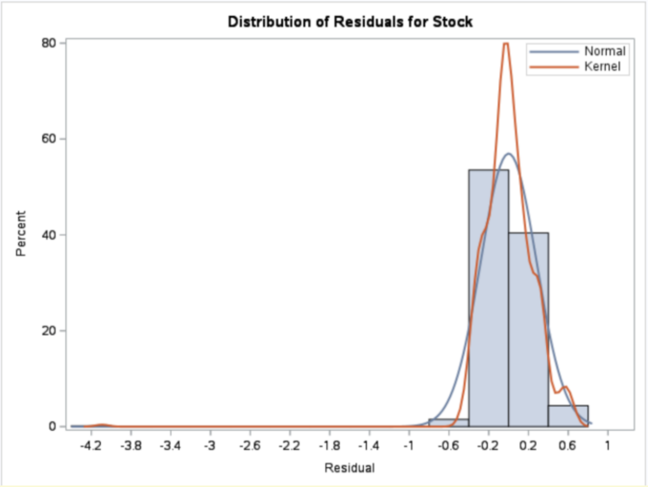
Pearson Correlation Coefficients, N = 594 Prob > |r| under H0: Rho=0							
	Stock	**Basket_index**	**EPS**	**P_E_ratio**	**Media_analytics_index**	**M1_money_supply_index**	**Top_10_GDP**
Stock	1.00000	0.72596 <.0001	0.81854 <.0001	0.55020 <.0001	0.45251 <.0001	-0.81701 <.0001	0.72561 <.0001
Basket_index	0.72596 <.0001	1.00000	0.64644 <.0001	0.57384 <.0001	0.41024 <.0001	-0.86450 <.0001	0.55789 <.0001
EPS	0.81854 <.0001	0.64644 <.0001	1.00000	0.53717 <.0001	0.36946 <.0001	-0.72042 <.0001	0.68957 <.0001
P_E_ratio	0.55020 <.0001	0.57384 <.0001	0.53717 <.0001	1.00000	0.53504 <.0001	-0.70295 <.0001	0.45114 <.0001
Media_analytics_index	0.45251 <.0001	0.41024 <.0001	0.36946 <.0001	0.53504 <.0001	1.00000	-0.44001 <.0001	0.16492 <.0001
M1_money_supply_index	-0.81701 <.0001	-0.86450 <.0001	-0.72042 <.0001	-0.70295 <.0001	-0.44001 <.0001	1.00000	-0.62650 <.0001
Top_10_GDP	0.72561 <.0001	0.55789 <.0001	0.68957 <.0001	0.45114 <.0001	0.16492 <.0001	-0.62650 <.0001	1.00000

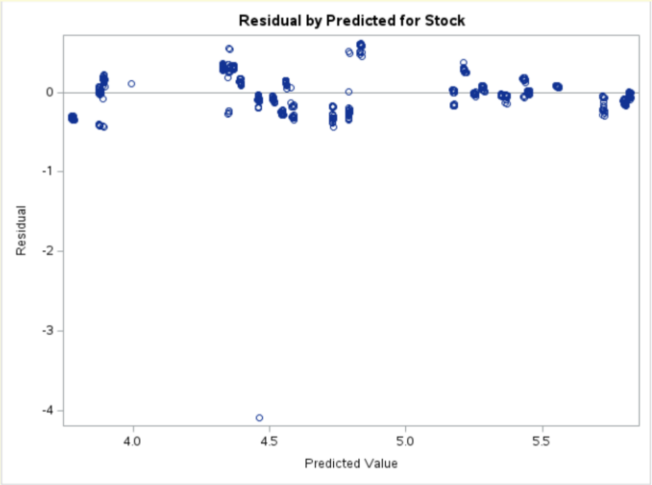
- Higher absolute values indicate stronger relationships

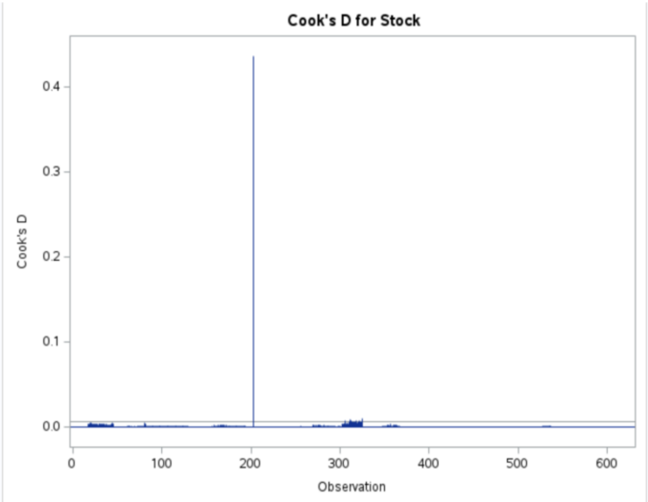
- This helps determine which variables may influence stock price

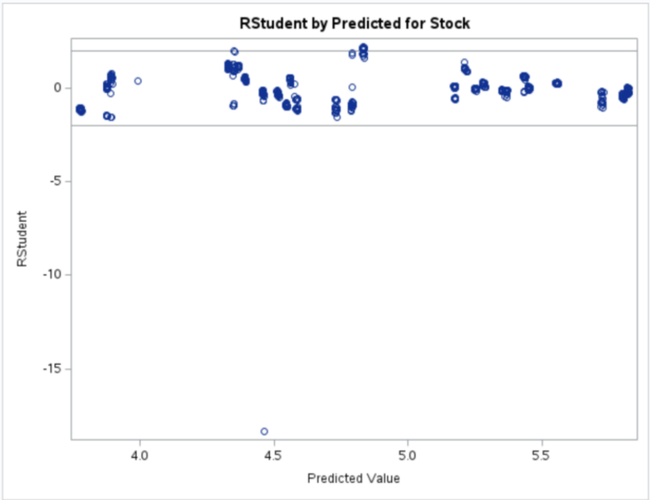
Regression Diagnostics

Output:









- Diagnostic plots evaluate model assumptions
- Residual plots assess homoscedasticity
- The histogram checks normality of residuals
- Cook's D identifies influential observations

MULTICOLLINEARITY - VIF

- Variance Inflation Factor (VIF) measures multicollinearity
- High VIF values indicate strong correlation between independent variables
- Multicollinearity can reduce model reliability

The output:

CODE LOG RESULTS OUTPUT DATA

Table of Contents
- Fit
  - Stock
    - Number of Observations
    - Analysis of Variance
    - Fit Statistics
    - Parameter Estimates
- Observation-wise Statistics

Root MSE	0.28162	R-Square	0.8197
Dependent Mean	4.88315	Adj R-Sq	0.8178
Coeff Var	5.76713		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.29821	1.38243	3.11	0.0020
Basket_index	1	-0.00098135	0.00319	-0.31	0.7584
EPS	1	1.30051	0.10761	12.09	<.0001
P_E_ratio	1	-0.06820	0.01208	-5.64	<.0001
Media_analytics_index	1	0.01924	0.00245	7.85	<.0001
M1_money_supply_index	1	-0.06664	0.00641	-10.39	<.0001
Top_10_GDP	1	0.44928	0.04626	9.71	<.0001

Durbin-Watson / Autocorrelation Test

- The Durbin-Watson statistic tests for autocorrelation in residuals
- A value near 2 indicates little to no autocorrelation

The output:

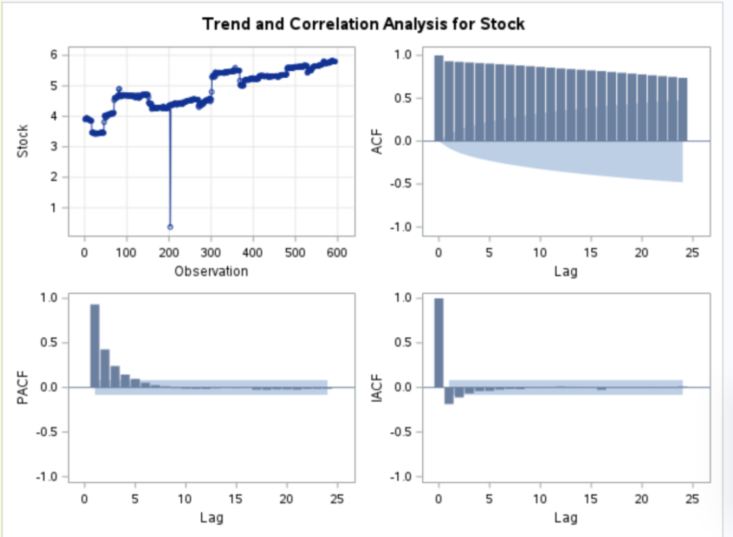
- Autocorrelation can affect regression validity in timeseries data

The REG Procedure
Model: MODEL1
Dependent Variable: Stock

Durbin-Watson D	0.738
Number of Observations	594
1st Order Autocorrelation	0.631

ARIMA – ACF & PACF

Output:



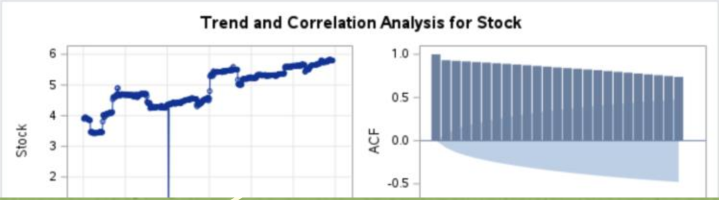
Boxplot of Stock Prices

The ARIMA Procedure

Name of Variable = Stock	
Mean of Working Series	4.883148
Standard Deviation	0.65929
Number of Observations	594

Autocorrelation Check for White Noise

To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	3006.97	6	<.0001	0.932	0.924	0.917	0.910	0.904	0.897
12	5755.13	12	<.0001	0.890	0.882	0.874	0.866	0.857	0.848
18	8224.08	18	<.0001	0.840	0.833	0.824	0.817	0.807	0.797
24	9999.99	24	<.0001	0.788	0.778	0.768	0.758	0.749	0.738



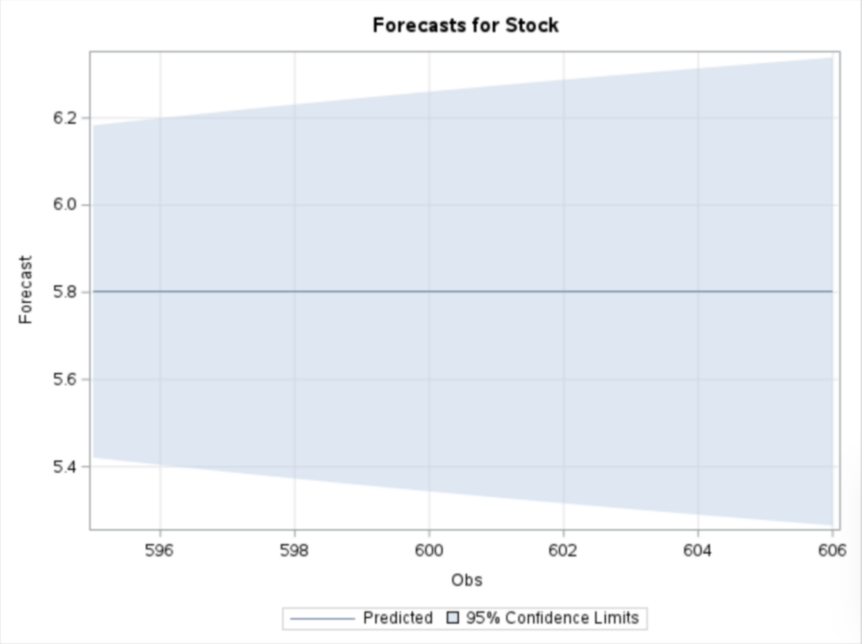
- ARIMA models use past values to forecast future values
- The ACF plot shows autocorrelation structure
- The PACF plot helps identify AR terms
- ARIMA is commonly used for time series forecasting

ARIMA Forecast Results

•The ARIMA model was estimated using historical stock data

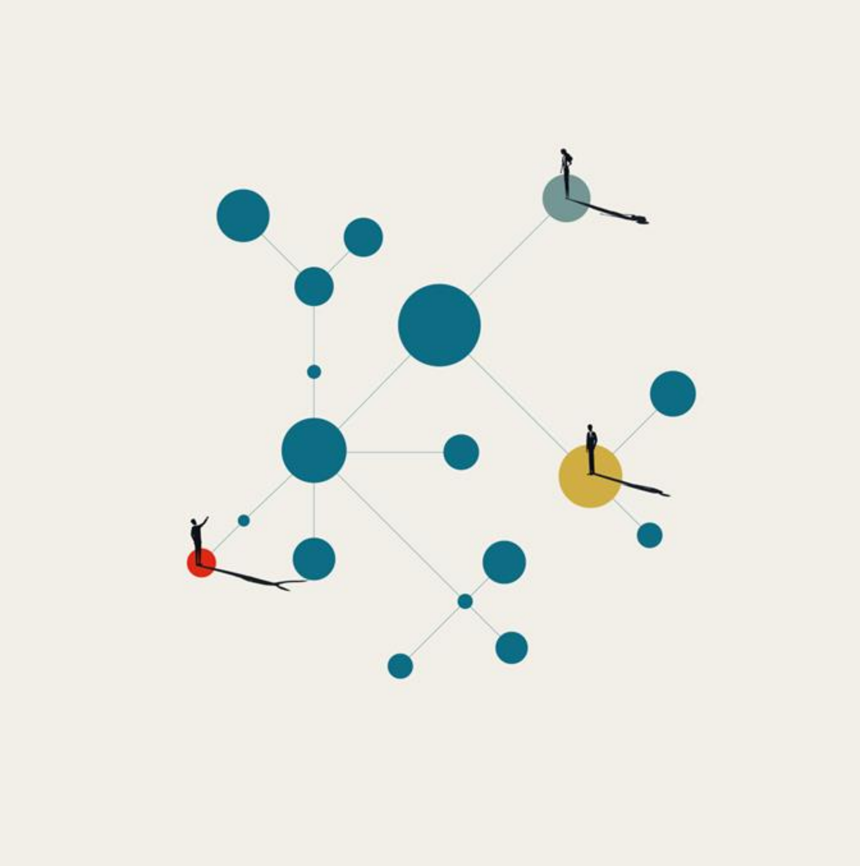
•Forecasts were generated for future periods

The output:



Forecasts for variable Stock				
Obs	**Forecast**	**Std Error**	**95% Confidence Limits**	
595	5.8020	0.1944	5.4210	6.1829
596	5.8020	0.2029	5.4043	6.1996
597	5.8020	0.2111	5.3883	6.2156
598	5.8020	0.2189	5.3729	6.2310
599	5.8020	0.2265	5.3580	6.2459
600	5.8020	0.2339	5.3436	6.2603
601	5.8020	0.2410	5.3296	6.2743
602	5.8020	0.2479	5.3161	6.2879
603	5.8020	0.2546	5.3029	6.3011
604	5.8020	0.2612	5.2900	6.3139
605	5.8020	0.2676	5.2775	6.3264
606	5.8020	0.2738	5.2652	6.3387

- The model provides projected stock values based onpast patterns



THANK YOU FOR YOUR TIME, DR. AHMETI